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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/123,186	03/17/2023	Yu Fang	SONA-0010-U01-C01-C06	9111

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EXAMINER	
ALGIBHAH, HAMZA N	

ART UNIT	PAPER NUMBER
2441	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/123,186	Applicant(s) Fang et al.	
	Examiner HAMZA N ALGIBHAH	Art Unit 2441	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 03/17/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-22 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-4 and 12-15 is/are rejected.

8) ☑ Claim(s) 5-11 and 16-22 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 03/17/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. ***Claims 1-22*** are pending.

2. ***Claims 1-4 and 12-15 are rejected.***

3. ***Claims 5-11 and 16-22 are objected.***

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

4. ***Claims 1-4 and 12-15 are rejected*** under 35 U.S.C. 102(a)(1) as being anticipated by Slusar (Pub. No.: US 2017/0228410 A1).

As per claim 1, Slusar discloses s vehicle comprising:

- ***a vehicle communication system*** (Slusar, Fig 3,) ***including:***

- ***a policy manager circuit configured to interpret a data collection policy including a trigger condition, a vehicle data identifier configured to identify vehicle data to be captured in response to a trigger event occurrence, and a trigger evaluation data identifier configured to identify trigger evaluation data to be captured in response to the trigger condition*** (Slusar, Fig 4, paragraph 0075-0076, 0079, wherein the on-board computer and/or processing system 313 may initiate data recording by a second data recording device (e.g., mobile computing device 314, dongle, etc.). In some arrangements, this step may be performed upon starting the vehicle 310. Alternatively, data recording by the second data recording device may be triggered at a

random time during a time interval coinciding with the starting and stopping (i.e., engine ignition and cutoff) of vehicle 310. In some instances, data recording may be initiated at a defined interval coinciding with the starting and stopping of vehicle 310 (i.e., 1 minute after the engine is started until the engine is cutoff, 5 minutes after the engine is started until the engine is cutoff, and the like). In other instances, data recording of the second data recording device may be triggered by a remote request from one of the first data recording device, on-board computer and/or processing system 313, data authenticity server 328, and the like. Additionally, in embodiments in which the second data recording device is mobile computing device 314, data recording may be triggered upon mobile computing device 314 receiving a text message, phone call, and/or push notification from an application (e.g., Facebook, Snapchat, WhatsApp, and the like) operating on mobile computing device 314. In other arrangements, data recording may be initiated after data communication has been established between on-board computer and/or processing system 313 and the first data recording device. In addition, on-board computer and/or processing system 313 may protect the data stream by using watermarking and/or encryption techniques discussed herein. As stated above, watermarking and/or encrypting the data stream may protect the data from being modified and/or may provide the ability to detect modification and/or corruption of the data. The watermark (e.g., data code, data key, or the like) may be inserted periodically, aperiodically, or randomly. The location, contents, and/or form of the watermark may be used in determining whether the data stream is authentic wherein the watermark can be (or at least include) the vehicle data identifier. Since Slusar discloses a trigger event (for example, starting and stopping of vehicle), trigger condition (for example, 5 minutes after the engine is started until the engine is cutoff) and to start capturing/recording data when the trigger even occur and the trigger condition is met, Slusar by inherence teaches a data collection policy and the component responsible for this policy can be

the policy manager circuit as claimed)***; and***

- ***an end point configured to provide a raw vehicle data stream including a trigger evaluation data stream and an identified vehicle data stream in response to the data collection policy*** (Slusar, Fig 4, paragraph 0078-0079, wherein on-board computer and/or processing system 313 may prepare raw data including first vehicle operational information and/or performance data received from vehicle operation sensors 211 and cameras/proximity sensors 212 for output to, for example, data recorder 316. The raw data may include vehicle operational information and/or performance data which may comprise engine temperature, engine RPM, vehicle velocity, vehicle acceleration, applied brake force, interior vehicle temperature, exterior vehicle temperature, window displacement, exterior vehicle barometric pressure, exhaust pressure, vehicle emissions, turbo blower pressure, turbo charger RPM, vehicle GPS location, and the like and formatting the raw data into a transmittable data stream. In addition, on-board computer and/or processing system 313 may format the data stream to protect the authenticity of the data comprised therein. For example, on-board computer and/or processing system 313 may protect the data stream by using watermarking. The watermark (e.g., data code, data key, or the like) may be inserted periodically, aperiodically, or randomly. Therefore, the stream includes an identified vehicle data stream as claimed).

As per claim 2, claim 1 is incorporated and Slusar further discloses

a filtering circuit configured to determine the trigger evaluation data stream of the raw vehicle data stream in response to trigger evaluation data identifier and provide the trigger evaluation data stream (Slusar, Fig 4, paragraph 0078-0079, wherein the raw data may include vehicle operational information and/or performance data which may comprise engine temperature, engine RPM, vehicle velocity, vehicle acceleration, applied brake force, interior

vehicle temperature, exterior vehicle temperature, window displacement, exterior vehicle barometric pressure, exhaust pressure, vehicle emissions, turbo blower pressure, turbo charger RPM, vehicle GPS location, and the like. The vehicle operational information and/or performance data may be further associated with the time period at which the data was recorded. In some instances, the raw data received by on-board computer and/or processing system 313 from vehicle operation sensors 211 and cameras/proximity sensors 212 may be encrypted according to the encryption techniques described herein. The preparation of raw data by on-board computer and/or processing system 313 may include processing operational and/or performance signals from sensors of vehicle 310 into raw data, determining the amount of raw data called for by, for example, data recorder 316 in accordance with the type of vehicle 310, driver rating, type of insurance policy, etc., determining the communication interface (e.g., data port, Bluetooth, Wi-Fi, or the like) through which to send the raw data to, for example, data recorder 316, and formatting the raw data into a transmittable data stream)*;*

As per claim 3, claim 2 is incorporated and Slusar further discloses

wherein the filtering circuit is configured to determine the identified vehicle data stream in response to the vehicle data identifier (Slusar, Fig 4, paragraph 0078-0079, wherein the data recorder **316**, mobile computing device **314**, and/or data authenticity server **328** may determine the presence or absence of the watermark in assessing the authenticity of the data when receiving the data stream from on-board computer and/or processing system **313**. For example, an absence of the watermark in the predetermined location and/or presence of an incorrect watermark may indicate that the data has been modified and/or corrupted rendering the data inauthentic)*;*

As per claim 4, claim 1 is incorporated and Slusar further discloses vehicle data processing circuit configured to perform at least one of sample the trigger evaluation data stream in response to a sampling parameter of the data collection policy, normalize a trigger evaluation data value format, or determine a trigger evaluation data aggregation parameter in response to a plurality of trigger conditions of the data collection policy (Zhang, Fig 4, paragraph 0056-0057, wherein on-board computer and/or processing system 313 may format the vehicle performance and/or operational data according to aspects described herein to protect the vehicle performance and/or operational data from modification and/or corruption. For example, the on-board computer and/or processing system 313 may encrypt and/or watermark the vehicle performance and/or operational data before outputting the data to data recorder 312)***;***

Claims 12-15 are rejected under the same rationale as claim 1-4.

Allowable Subject Matter

Claims ***5-11 and 16-22 are*** objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached 7:30 am - 3:30 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Wing Chan can be reached on (571) 272-7493. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/HAMZA N ALGIBHAH/
Primary Examiner, Art Unit 2441